

1. CSS GRID — Complete Properties

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Property / Method	Kya kaam karta hai	Example	Deep Explanation
<code>display: grid</code>	Element ko Grid Container banata hai	<code>display: grid;</code>	Iske direct children grid items ban jaate hain. Grid rows aur columns ke through 2D layout create karta hai.
<code>grid-template-rows</code>	Rows ki size/number define karta hai	<code>grid-template-rows: 100px 200px 100px;</code>	Har value ek row ka track size define karti hai.
<code>grid-template-columns</code>	Columns ki size/number define karta hai	<code>grid-template-columns: 200px 1fr 200px;</code>	Har value ek column ka track size define karti hai.
<code>grid-template-areas</code>	Named grid areas banata hai	<code>grid-template-areas: "header header" "side main";</code>	Layout ko names dekar readable way me design kar sakte ho.
<code>grid-template</code>	Rows + columns + areas ka shorthand	<code>grid-template: 100px 1fr / 200px 1fr;</code>	<code>grid-template-rows</code> , <code>grid-template-columns</code> aur optionally areas ko shorthand me define karta hai.

			shorthand me define karta hai.
<code>grid-auto-rows</code>	Automatically created rows ki size	<code>grid-auto-rows: 100px;</code>	Agar explicit rows khatam ho jaayein aur naye items ke liye implicit rows banein, unki size control karta hai.
<code>grid-auto-columns</code>	Automatically created columns ki size	<code>grid-auto-columns: 150px;</code>	Implicit columns ki size control karta hai.
<code>grid-auto-flow</code>	Auto-placement direction	<code>grid-auto-flow: row;</code>	Browser automatically grid items ko kis direction me place karega, decide karta hai.
<code>grid-row</code>	Item ko row lines par place karta hai	<code>grid-row: 1 / 3;</code>	Item row line 1 se start hokar line 3 tak jayega, yani 2 row tracks cover karega.
<code>grid-row-start</code>	Row starting line	<code>grid-row-start: 1;</code>	Grid item kis horizontal grid line se start hoga.
<code>grid-row-end</code>	Row ending line	<code>grid-row-end: 3;</code>	Item kis horizontal grid line par end hoga
<code>grid-row-end</code>	Row ending line	<code>grid-row-end: 3;</code>	Item kis horizontal grid line par end hoga.
<code>grid-column</code>	Item ko columns me place karta hai	<code>grid-column: 1 / 3;</code>	Column line 1 se 3 tak item span karega.
<code>grid-column-start</code>	Column starting line	<code>grid-column-start: 1;</code>	Item kis vertical grid line se start hoga.
<code>grid-column-end</code>	Column ending line	<code>grid-column-end: 3;</code>	Item kis vertical grid line par end hoga.
<code>grid-area</code>	Area name ya row/column placement	<code>grid-area: header;</code>	Named area assign kar sakta hai ya <code>row-start / column-start / row-end / column-end</code> shorthand use kar sakta hai.
<code>gap</code>	Rows + columns ke beech gap	<code>gap: 20px;</code>	Grid items ke beech spacing set karta hai.
<code>row-gap</code>	Rows ke beech gap	<code>row-gap: 20px;</code>	Sirf row direction ka gap.

<code>row-gap</code>	Rows ke beech gap	<code>row-gap: 20px;</code>	Siri row direction ka gap.	↑ Share
<code>column-gap</code>	Columns ke beech gap	<code>column-gap: 30px;</code>	Sirf column direction ka gap.	
<code>justify-items</code>	Grid cell ke andar horizontal/inline alignment	<code>justify-items: center;</code>	Har grid item ko uske own grid area/cell ke andar inline axis par align karta hai.	
<code>align-items</code>	Grid cell ke andar vertical/block alignment	<code>align-items: center;</code>	Har grid item ko uske cell ke andar block/cross axis par align karta hai.	
<code>place-items</code>	<code>align-items</code> + <code>justify-items</code> shorthand	<code>place-items: center;</code>	Dono directions me same alignment chahiye ho to useful.	
<code>justify-content</code>	Puri grid ko container ke andar inline axis par align	<code>justify-content: center;</code>	Grid ki total width container se chhoti ho to remaining space distribute/position karta hai.	
<code>align-content</code>	Puri grid ko container ke andar block axis par align	<code>align-content: center;</code>	Grid ki total height container se chhoti ho to grid ko vertically position/distribute karta hai.	
<code>align-content</code>	Puri grid ko container ke andar block axis par align	<code>align-content: center;</code>	Grid ki total height container se chhoti ho to grid ko vertically position/distribute karta hai.	↑ Share
<code>place-content</code>	<code>align-content</code> + <code>justify-content</code> shorthand	<code>place-content: center;</code>	Puri grid ko dono directions me align karta hai.	
<code>justify-self</code>	Individual item ki horizontal alignment	<code>justify-self: end;</code>	Parent ke <code>justify-items</code> ko individual item ke liye override karta hai.	
<code>align-self</code>	Individual item ki vertical alignment	<code>align-self: center;</code>	Parent ke <code>align-items</code> ko individual item ke liye override karta hai.	
<code>place-self</code>	<code>align-self</code> + <code>justify-self</code> shorthand	<code>place-self: center;</code>	Individual grid item ko dono directions me align karta hai.	

2. grid-template-rows — Complete Methods

Method / Value	Kya karta hai	Example	Deep Explanation
px	Fixed row size	<code>grid-template-rows: 100px 200px;</code>	Row ki fixed height.
%	Container ke percentage ke according	<code>grid-template-rows: 20% 80%;</code>	Available grid container size ke percentage se track size.
auto	Content/available space ke according	<code>grid-template-rows: auto auto;</code>	Browser content aur available space ke according size calculate karta hai.
fr	Available space ka fraction	<code>grid-template-rows: 1fr 2fr;</code>	Available free space ko 1:2 ratio me distribute karta hai.
minmax()	Minimum + maximum size	<code>minmax(100px, 1fr)</code>	Track 100px se chhota nahi hoga, available space me 1fr tak grow kar sakta hai.
min-content	Content ki minimum required size	<code>grid-template-rows: min-content;</code>	Content ko minimum possible size tak shrink karta hai.
max-content	Content ki maximum preferred size	<code>grid-template-rows: max-content;</code>	Content ko generally ek line/preferred maximum size tak accommodate karta hai.
fit-content()	Maximum limit ke saath content-based size	<code>fit-content(300px)</code>	Track content ke according grow karta hai but specified limit cross nahi karta.
repeat()	Repeated tracks	<code>repeat(3, 100px)</code>	Same track ko baar-baar likhne ki zarurat nahi
subgrid	Parent grid tracks inherit	<code>grid-template-rows: subgrid;</code>	Nested grid parent ki row tracks ko use kar sakta hai.

3. grid-template-columns — Complete Methods



Method / Value	Kya karta hai	Example	Deep Explanation	
px	Fixed column	100px 200px 100px	Fixed width columns.	
%	Percentage width	25% 75%	Container ke available inline size ke percentage ke according.	
auto	Automatic sizing	auto auto auto	Content aur available space ke according size.	
fr	Fractional available space	1fr 1fr 1fr	Remaining space ko equal columns me divide karta hai.	
2fr	Double fraction	1fr 2fr	Second column ko available free space me first ke comparison me 2 fraction mil sakte hain.	
minmax()	Minimum/maximum track	minmax(200px, 1fr) 	Responsive grid ke liye extremely useful.	

<code>minmax()</code>	minimum/maximum track	<code>minmax(200px, 1fr)</code>	responsive grid ke nye extremely useful.
<code>min-content</code>	Minimum content size	<code>min-content</code>	Content ko minimum intrinsic size tak fit karta hai.
<code>max-content</code>	Maximum content size	<code>max-content</code>	Content ki preferred maximum intrinsic size.
<code>fit-content()</code>	Content + maximum limit	<code>fit-content(300px)</code>	Content ke according size but max limit ke saath.
<code>repeat()</code>	Tracks repeat	<code>repeat(3, 1fr)</code>	3 equal columns.
<code>auto-fill</code>	Available tracks fill karta hai	<code>repeat(auto-fill, minmax(200px, 1fr))</code>	Available width me jitne tracks fit ho sakein unhe create karta hai.
<code>auto-fit</code>	Empty tracks collapse kar sakta hai	<code>repeat(auto-fit, minmax(200px, 1fr))</code>	Available items ke according tracks expand ho sakte hain.
<code>subgrid</code>	Parent ke tracks use	 <code>grid-template-</code>	Nested grid parent ki column

subgrid

Parent ke tracks use karta hai

grid-template-columns: subgrid;

Nested grid parent ki column structure follow kar sakta hai.

4. `fr` Unit — Deep Explanation

Code

Meaning



`grid-template-columns: 1fr 1fr;`

Available free space ko 2 equal parts me divide

`grid-template-columns: 1fr 2fr;`

Free space ka ratio 1:2

`grid-template-columns: 1fr 1fr 1fr;`

3 equal flexible columns

`grid-template-columns: 2fr 1fr;`

First ko second ke comparison me 2 fraction, second ko 1

`grid-template-columns: 200px 1fr;`

First fixed 200px, remaining space second ko

`grid-template-columns: 200px 1fr 1fr;`

First  d, remaining space two equal parts

5. `repeat()` — Complete Methods

Syntax	Meaning
<code>repeat(3, 1fr)</code>	3 equal <code>1fr</code> columns
<code>repeat(4, 100px)</code>	4 columns, each 100px
<code>repeat(3, 100px 200px)</code>	Pattern <code>100px 200px</code> three times
<code>repeat(3, 1fr 2fr)</code>	Pattern <code>1fr 2fr</code> three times
<code>repeat(auto-fit, minmax(200px, 1fr))</code>	Responsive columns
<code>repeat(auto-fill, minmax(200px, 1fr))</code>	Available space me maximum fitting tracks

6. minmax() — Deep

Code	Kya hoga	
<code>minmax(100px, 1fr)</code>	Track minimum 100px; free space available ho to grow	
<code>minmax(200px, 300px)</code>	Track 200–300px ke beech	
<code>minmax(150px, max-content)</code>	Minimum 150px, maximum content-based	
<code>repeat(auto-fit, minmax(200px, 1fr))</code>	Responsive card grid ka extremely common pattern	
Part	Meaning	
<code>minmax(200px, 1fr)</code> ka first part	Minimum	
<code>minmax(200px, 1fr)</code> ka second part	Maximum	
<code>1fr</code>	Available free space me flexible growth	

7. Grid Gap

Property	Example	Kya karta hai
<code>gap</code>	<code>gap: 20px;</code>	Row + column dono gap
<code>row-gap</code>	<code>row-gap: 10px;</code>	Rows ke beech
<code>column-gap</code>	<code>column-gap: 20px;</code>	Columns ke beech
<code>gap: 10px 20px</code>	Row=10px, Column=20px	Two-value shorthand

8. Grid Item Placement — Complete

Property / Method	Example	Deep Explanation
<code>grid-column-start</code>	<code>grid-column-start: 1;</code>	Item column line 1 se start.
<code>grid-column-end</code>	<code>grid-column-end: 3;</code>	Column line 3 par end.
<code>grid-column</code>	<code>grid-column: 1 / 3;</code>	Start/end ko shorthand me define.
<code>grid-row-start</code>	<code>grid-row-start: 1;</code>	Row line 1 se start.
<code>grid-row-end</code>	<code>grid-row-end: 3;</code>	Row line 3 par end.
<code>grid-row</code>	<code>grid-row: 1 / 3;</code>	Row placement shorthand.
<code>span</code>	<code>grid-column: 1 / span 2;</code>	Starting line se 2 columns occupy karega.
Negative line	<code>grid-column: 1 / -1;</code>	First line se last grid line tak.
<code>grid-area</code>	<code>grid-area: header;</code>	Named area me item place.

9. Grid Placement Examples

Code	Result
<code>grid-column: 1 / 3;</code>	Item columns 1–2 occupy karega
<code>grid-column: 1 / span 2;</code>	Starting line 1 se 2 columns
<code>grid-column: 2 / 5;</code>	Columns 2–4
<code>grid-column: 1 / -1;</code>	First se last line tak full width
<code>grid-row: 1 / 3;</code>	Rows 1–2
<code>grid-row: 2 / span 3;</code>	Row 2 se 3 rows
<code>grid-area: header;</code>	Named <code>header</code> area me place

10. Grid Alignment — Bahut Important Difference

Property	Kis cheez ko align karta hai	Axis	Example	
<code>justify-items</code>	Har grid item ko uske cell ke andar	Inline/horizontal	<code>justify-items: center;</code>	
<code>align-items</code>	Har grid item ko cell ke andar	Block/vertical	<code>align-items: center;</code>	
<code>place-items</code>	<code>justify-items</code> + <code>align-items</code>	Both	<code>place-items: center;</code>	
<code>justify-self</code>	Ek individual item	Inline/horizontal	<code>justify-self: end;</code>	
<code>align-self</code>	Ek individual item	Block/vertical	<code>align-self: center;</code>	
<code>place-self</code>	<code>justify-self</code> + <code>align-self</code>	Both	<code>place-self: center;</code>	
				
<code>justify-content</code>	Puri grid	Inline/horizontal	<code>justify-content: center;</code>	
<code>align-content</code>	Puri grid	Block/vertical	<code>align-content: center;</code>	
<code>place-content</code>	<code>justify-content</code> + <code>align-content</code>	Both	<code>place-content: center;</code>	

11. Grid Alignment Values

Value	Meaning
start	Start side
end	End side
center	Center
stretch	Available area me stretch
space-between	Space between tracks/items
space-around	Around space
space-evenly	Equal spaces
normal	Default behavior
baseline	Text baseline alignment



safe center

Overflow situation me safer center alignment

unsafe center

Center alignment ko prefer/force karne wala form

12. grid-auto-flow — Complete

Value	Kya karta hai	Example
row	Items rows me automatically place	<code>grid-auto-flow: row;</code>
column	Items columns me place	<code>grid-auto-flow: column;</code>
dense	Earlier empty spaces ko later items se fill karne ki koshish	<code>grid-auto-flow: dense;</code>
row dense	Row placement + dense packing	<code>grid-auto-flow: row dense;</code>
column dense	Column placement + dense packing	<code>grid-auto-flow: column dense;</code>



13. Grid Template Areas

Property	Example	Deep Explanation
<code>grid-template-areas</code>	<code>grid-template-areas: "header header" "side main";</code>	Grid ko named regions me divide karta hai.
<code>grid-area</code>	<code>grid-area: header;</code>	Item ko named region assign karta hai.

HTML

CSS

<code>.header</code>	<code>grid-area: header;</code>
<code>.sidebar</code>	<code>grid-area: sidebar;</code>
<code>.main</code>	<code>grid-area: main;</code>
<code>.footer</code>	<code>grid-area: footer;</code>

14. Grid Container vs Grid Item

Grid Container

`display: grid`

`grid-template-columns`

`grid-template-rows`

`grid-template-areas`

`grid-template`

`grid-auto-columns`

`grid-auto-rows`

`grid-auto-flow`

`gap`

Grid Item

`grid-column`

`grid-column-start`

`grid-column-end`

`grid-row`

`grid-row-start`

`grid-row-end`

`grid-area`

`justify-self`

`align-self`



Flexbox Properties	
gap	align-self
row-gap	place-self
column-gap	—
justify-items	—
align-items	—
place-items	—
justify-content	—
align-content	—
place-content	—

15. CSS Grid ke Important Responsive Methods

Method	Example	Deep Explanation	
auto-fit	<code>repeat(auto-fit,minmax(200px,1fr))</code>	Available width ke according cards ki count adjust karta hai.	
auto-fill	<code>repeat(auto-fill,minmax(200px,1fr))</code>	Possible tracks ko fill karne ki koshish karta hai, even when some tracks empty ho sakte hain.	
minmax()	<code>minmax(200px,1fr)</code>	Minimum card width maintain karte hue responsive growth.	
fr	<code>1fr</code>	Remaining space ko flexible way me distribute.	
%	<code>25% 75%</code>	Percentage-based responsive sizing.	
auto	<code>auto 1fr</code>	Content/available space based sizing.	



16. CSS GRID — Important Functions

Function	Kya karta hai	Example
<code>repeat()</code>	Track repetition	<code>repeat(3, 1fr)</code>
<code>minmax()</code>	Minimum + maximum	<code>minmax(200px, 1fr)</code>
<code>fit-content()</code>	Content-based size with limit	<code>fit-content(300px)</code>
<code>calc()</code>	Calculation	<code>calc(50% - 20px)</code>
<code>min()</code>	Minimum value	<code>min(400px, 100%)</code>
<code>max()</code>	Maximum value	<code>max(200px, 30%)</code>
<code>clamp()</code>	Min + preferred + max	<code>clamp(200px, 30vw, 500px)</code>

17. CSS ANIMATION — Complete Properties

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Animation topic me `animation-name`, `animation-duration`, `animation-timing-function`, `animation-delay`, `animation-iteration-count` aur `animation-direction` jaise properties use hoti hain.  3rd css new (1)

Property	Kya karta hai	Example	Deep Explanation 
<code>animation-name</code>	Kaunsa <code>@keyframes</code> animation chalega	<code>animation-name: tarun;</code>	<code>@keyframes tarun</code> ke naam ko element se connect karta hai.
<code>animation-duration</code>	Animation kitni der chalegi	<code>animation-duration: 2s;</code>	Ek iteration ko complete karne ka time.
<code>animation-timing-function</code>	Animation ki speed curve	<code>animation-timing-function: ease-in;</code>	Animation start/middle/end me kitni fast/slow hogi.
<code>animation-delay</code>	Start se pehle wait	<code>animation-delay: 1s;</code>	Animation 1 second delay ke baad start hogi.
<code>animation-iteration-count</code>	Kitni baar animation repeat 	<code>animation-iteration-count: infinite;</code>	1, 2, 3, infinite etc.

<code>animation-iteration-count</code>	Kitni baar animation repeat	<code>animation-iteration-count: infinite;</code>	1, 2, 3, infinite etc Share
<code>animation-direction</code>	Animation kis direction me chale	<code>animation-direction: alternate;</code>	Forward/reverse/alternate behavior control.
<code>animation-fill-mode</code>	Start/end ke baad styles retain	<code>animation-fill-mode: forwards;</code>	Animation finish hone ke baad final keyframe style retain kar sakta hai.
<code>animation-play-state</code>	Animation play/pause	<code>animation-play-state: paused;</code>	Running animation ko pause/resume kar sakte ho.
<code>animation-timeline</code>	Animation ko timeline se link	<code>animation-timeline: scroll();</code>	Modern CSS me animation ko scroll/view progress se control kar sakte ho.
<code>animation-range</code>	Timeline animation ka active range	<code>animation-range: entry 0% cover 50%;</code>	Scroll/view timeline animations me animation kis range me active hogi control karta hai.
<code>animation-</code>	Multiple animations ke	<code>animation-composition:</code>	Multiple animations same
<code>animation-composition</code>	Multiple animations ke property effects combine kaise hon	<code>animation-composition: add;</code>	Multiple animations same property modify karein to composition behavior define karta hai.
<code>animation</code>	Animation shorthand	<code>animation: tarun 2s ease 1s infinite alternate;</code>	Multiple animation properties ko ek declaration me set karta hai.

18. animation-direction — Complete Values

File me `alternate`, `normal`, `reverse`, `alternate-reverse` values shown hain.

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Value	Kya karta hai
<code>normal</code>	0% → 100%, har iteration same forward direction
<code>reverse</code>	100% → 0%, reverse direction
<code>alternate</code>	Forward → Reverse → Forward → Reverse
<code>alternate-reverse</code>	Reverse → Forward → Reverse → Forward

19. animation-iteration-count

Value	Meaning
1	Ek baar
2	Do baar
3	Teen baar
5	Five times
infinite	Continuously repeat
Example	Result
<code>animation-iteration-count: 1;</code>	Once
<code>animation-iteration-count: 3;</code>	3 iterations
<code>animation-iteration-count: infinite;</code>	↓ Unlimited

20. animation-timing-function

Value / Method	Kya karta hai
<code>linear</code>	Constant speed
<code>ease</code>	Slow → fast → slow
<code>ease-in</code>	Slow start
<code>ease-out</code>	Slow end
<code>ease-in-out</code>	Slow start + slow end
<code>step-start</code>	Immediately first step
<code>step-end</code>	Change at end
<code>steps(5)</code>	5 discrete steps
<code>cubic-bezier()</code>	Custom timing curve

21. Animation Delay

Code	Result
<code>animation-delay: 0s;</code>	Immediately
<code>animation-delay: 1s;</code>	1 second wait
<code>animation-delay: 2s;</code>	2 second wait
<code>animation-delay: -1s;</code>	Animation timeline ko effectively 1 second ahead se start kar sakta hai

22. animation-fill-mode

Value	Kya karta hai
none	Animation se pehle/baad keyframe styles retain nahi
forwards	Animation ke end state ko retain
backwards	Delay period me first relevant keyframe styles apply
both	forwards + backwards

23. animation-play-state

Value	Kya karta hai
running	Animation chalti rahegi
paused	Animation pause

Example	Use
animation-play-state: paused;	Hover par animation pause
animation-play-state: running;	Hover hataane par resume

24. @keyframes — Complete Methods

Method	Example	Deep Explanation	
from	<code>from { opacity: 0; }</code>	Animation ki starting state.	
to	<code>to { opacity: 1; }</code>	Animation ki ending state.	
Percentage	<code>0% { ... }</code>	Exact timeline position.	
25%	<code>25% { ... }</code>	Animation ke 25% point par style.	
50%	<code>50% { ... }</code>	Middle state.	
75%	<code>75% { ... }</code>	75% state.	
100%	<code>100% { ... }</code>	Final state.	

25. Keyframes me kya-kya animate kar sakte ho?

Property Category	Examples	Use
Transform	<code>transform</code>	Move, scale, rotate
Color	<code>color</code>	Text color animation
Background	<code>background-color</code>	Background color change
Size	<code>width</code> , <code>height</code>	Size animation
Position	<code>top</code> , <code>left</code> , etc.	Position changes
Opacity	<code>opacity</code>	Fade in/out
Shadow	<code>box-shadow</code>	Glow/elevation
Border	<code>border-color</code> , <code>border-radius</code>	Border effects
Typography	<code>font-size</code> , <code>letter-spacing</code>	Text effects



Filter	<code>filter</code>	Blur/brightness etc.
Clip	<code>clip-path</code>	Shape animation

26. Animation Shorthand — Deep

Code	Meaning
<code>animation: tarun 2s;</code>	Name + duration
<code>animation: tarun 2s ease;</code>	Name + duration + timing
<code>animation: tarun 2s ease 1s;</code>	+ delay
<code>animation: tarun 2s ease 1s infinite;</code>	+ iteration count
<code>animation: tarun 2s ease 1s infinite alternate;</code>	+ direction
<code>animation: tarun 2s ease 1s infinite alternate forwards;</code>	+ fill mode

```
animation: tarun 2s ease-in 1s infinite
alternate;
```

Complete common shorthand

27. Multiple Animations

Code	Meaning
<code>animation: move 2s, rotate 3s;</code>	Do animations simultaneously
<code>animation-name: move, rotate;</code>	Multiple animation names
<code>animation-duration: 2s, 3s;</code>	Corresponding durations
<code>animation-delay: 0s, 1s;</code>	Different delays

28. CSS Animation vs Transition

Feature	Transition	Animation
Trigger	Usually state change, e.g. <code>:hover</code>	Automatically run bhi ho sakti hai
Keyframes	✗	✓
<code>@keyframes</code>	✗	✓
Multiple stages	Limited	✓ <code>0%</code> , <code>25%</code> , <code>50%</code> , etc.
Infinite loop	✗ Normally	✓
Direction control	✗	✓
Delay	✓	✓
Timing function	✓	✓
Play/pause	✗ Direct animation control nahi	✓ <code>animation-play-state</code>

29. Responsive CSS / Media Query — Complete

Media Query topic ka purpose screen width ke according CSS change karna hai; file me `min-width`, `max-width` aur combined conditions ke examples diye gaye hain.

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Syntax / Feature	Kya karta hai	Example 
<code>@media</code>	Conditional CSS apply	<code>@media (max-width:600px){...}</code>
<code>min-width</code>	Minimum viewport width	<code>@media (min-width:600px)</code>
<code>max-width</code>	Maximum viewport width	<code>@media (max-width:600px)</code>
<code>min-height</code>	Minimum viewport height	<code>@media (min-height:700px)</code>
<code>max-height</code>	Maximum viewport height	<code>@media (max-height:700px)</code>
<code>width</code>	Exact/constraint width condition	<code>@media (width:600px)</code>
<code>height</code>	Exact/constraint height condition 	<code>@media (height:800px)</code>

height	Exact/constraint height condition	@media (height:800px)
orientation	Portrait/landscape	@media (orientation: landscape)
aspect-ratio	Viewport aspect ratio	@media (aspect-ratio: 16/9)
resolution	Display resolution	@media (min-resolution: 2dppx)
scan	Display scan type	@media (scan: progressive)
color	Color support	@media (color)
color-index	Color index capability	@media (min-color-index: 256)
monochrome	Monochrome display capability	@media (monochrome)
hover	Primary input device capability	@media (hover: hover)

<code>pointer</code>	Pointer accuracy	<code>@media (pointer: fine)</code>
<code>any-hover</code>	Any available input hover support	<code>@media (any-hover: hover)</code>
<code>any-pointer</code>	Any available pointer capability	<code>@media (any-pointer: coarse)</code>
<code>prefers-color-scheme</code>	User light/dark preference	<code>@media (prefers-color-scheme: dark)</code>
<code>prefers-reduced-motion</code>	User motion preference	<code>@media (prefers-reduced-motion: reduce)</code>
<code>forced-colors</code>	Forced color mode	<code>@media (forced-colors: active)</code>

30. `min-width` vs `max-width`

Query	Meaning
<code>@media (min-width: 600px)</code>	600px ya usse bada viewport
<code>@media (max-width: 600px)</code>	600px ya usse chhota viewport
<code>@media (min-width: 400px) and (max-width: 900px)</code>	400–900px range
<code>@media (min-width: 901px)</code>	901px and above
<code>@media (max-width: 399px)</code>	399px and below

31. Media Query Operators

Operator	Kya karta hai	Example	
and	Dono conditions true honi chahiye	<code>(min-width:400px) and (max-width:900px)</code>	
or	Kisi ek condition ko satisfy karna	Modern media syntax me comma/list use commonly equivalent OR behavior deta hai	
,	Multiple media queries ko OR karta hai	<code>@media (max-width:500px), (orientation:portrait)</code>	
not	Condition ko negate karta hai	<code>@media not (hover: hover)</code>	

32. Media Query — Multiple Range Examples

Requirement	CSS
600px से ऊपर	<code>@media (min-width: 600px)</code>
600px से नीचे	<code>@media (max-width: 600px)</code>
400–900px	<code>@media (min-width: 400px) and (max-width: 900px)</code>
400px से नीचे OR 900px से ऊपर	<code>@media (max-width: 400px), (min-width: 900px)</code>
Landscape	<code>@media (orientation: landscape)</code>
Portrait	<code>@media (orientation: portrait)</code>
Dark mode	<code>@media (prefers-color-scheme: dark)</code>
Light mode	<code>@media (prefers-color-scheme: light)</code>
Reduced motion	<code>@media (prefers-reduced-motion: reduce)</code>

33. Media Query — Input Device Features

Feature	Values	Deep Explanation
<code>hover</code>	<code>hover</code> , <code>none</code>	Primary input device hover support करता है या नहीं।
<code>pointer</code>	<code>none</code> , <code>coarse</code> , <code>fine</code>	Pointer कितना accurate है।
<code>any-hover</code>	<code>hover</code> , <code>none</code>	Available inputs में कोई hover-capable device है या नहीं।
<code>any-pointer</code>	<code>none</code> , <code>coarse</code> , <code>fine</code>	Available pointers में capability क्या है।

34. Media Query — User Preferences

Feature	Values	Kya karta hai	
<code>prefers-color-scheme</code>	<code>light</code> , <code>dark</code>	User ke OS/browser theme preference ke according CSS	
<code>prefers-reduced-motion</code>	<code>reduce</code> , <code>no-preference</code>	User animation/motion ko reduce karna chahta hai ya nahi	
<code>prefers-contrast</code>	Preference values	User ki contrast preference ke according styling	
<code>forced-colors</code>	<code>active</code> , <code>none</code>	System forced-color mode detect	
<code>inverted-colors</code>	<code>inverted</code> , <code>none</code>	Inverted color preference/state	

35. Media Query — Screen/Display Features

Feature	Kya check karta hai	Example	
<code>width</code>	Viewport width	<code>(min-width: 600px)</code>	
<code>height</code>	Viewport height	<code>(min-height: 700px)</code>	
<code>aspect-ratio</code>	Width/height ratio	<code>(aspect-ratio: 16/9)</code>	
<code>orientation</code>	Portrait/landscape	<code>(orientation: portrait)</code>	
<code>resolution</code>	Display resolution	<code>(min-resolution: 2dppx)</code>	
<code>color</code>	Color display support	<code>(color)</code>	
<code>monochrome</code>	Monochrome display	<code>(monochrome)</code>	
<code>display-mode</code>	App/browser display mode	<code>(display-mode: standalone)</code>	

36. Responsive Design ke Units

Unit	Kiske relative hai	Example	Use	
%	Parent/container	<code>width: 50%;</code>	Fluid sizing	
vw	Viewport width	<code>width: 50vw;</code>	Screen-width based	
vh	Viewport height	<code>height: 100vh;</code>	Screen-height based	
vmin	Smaller viewport dimension	<code>10vmin</code>	Responsive proportional sizing	
vmax	Larger viewport dimension	<code>10vmax</code>	Responsive sizing	
rem	Root font size	<code>2rem</code>	Scalable typography/spacing	
em	Current/parent font context	<code>2em</code>	Component-relative sizing	
ch	Character width metric	<code>60ch</code>	Text line width	

37. Responsive Grid — Best Methods

Technique	Example	Deep Explanation	
Fixed breakpoints	<code>@media (max-width:600px)</code>	Specific screen size par layout change I	
<code>auto-fit</code>	<code>repeat(auto-fit,minmax(200px,1fr))</code>	Grid automatically columns adjust karta hai	
<code>auto-fill</code>	<code>repeat(auto-fill,minmax(200px,1fr))</code>	Available tracks fill karne ki koshish I	
<code>minmax()</code>	<code>minmax(200px,1fr)</code>	Card ki minimum width preserve.	
<code>clamp()</code>	<code>width:clamp(200px,30vw,400px)</code>	Smooth responsive size.	
<code>%</code>	<code>grid-template-columns: 30% 70%</code>	Container-based responsive layout.	
<code>fr</code>	<code>1fr 2fr</code>	Available space proportional distribution.	

38. CSS Animation + Responsive Media Query

Requirement	CSS Method
Mobile par animation kam	<code>@media (max-width:600px)</code>
User motion reduce chahta hai	<code>@media (prefers-reduced-motion: reduce)</code>
Animation completely disable	<code>animation: none;</code>
Animation slow karna	<code>animation-duration: 3s;</code>
Animation pause	<code>animation-play-state: paused;</code>
Mobile par smaller transform	Media query ke andar <code>transform</code> change
Desktop par animation	<code>@media (min-width:900px)</code>

39. `prefers-reduced-motion` — Important

Code	Kya karta hai
<code>@media (prefers-reduced-motion: reduce) { * { animation: none; transition: none; } }</code>	User ne motion reduce preference set ki ho to unnecessary animation/transition reduce/disable kar sakte ho
<code>animation: none</code>	Animation disable
<code>transition: none</code>	Transition disable

40. Grid + Media Query Combined

Screen	Grid
Desktop	<code>grid-template-columns: repeat(4,1fr)</code>
Tablet	<code>grid-template-columns: repeat(2,1fr)</code>
Mobile	<code>grid-template-columns: 1fr</code>

CSS Concept	Deep Meaning	
Desktop	More horizontal space → more columns	
Tablet	Medium space → fewer columns	
Mobile	Narrow space → one column	
<code>gap</code>	Cards ke beech consistent spacing	
<code>minmax()</code>	Cards ko too-small hone se prevent kar sakta hai	
<code>auto-fit</code>	Breakpoint ke bina automatic responsive behavior de sakta hai	

41. Grid + Animation + Media Query — Combined Example Concepts

Topic	Properties
Grid layout	<code>display:grid</code>
Columns	<code>grid-template-columns</code>
Rows	<code>grid-template-rows</code>
Responsive columns	<code>repeat()</code> , <code>auto-fit</code> , <code>auto-fill</code>
Minimum card size	<code>minmax()</code>
Spacing	<code>gap</code> , <code>row-gap</code> , <code>column-gap</code>
Card placement	<code>grid-column</code> , <code>grid-row</code> , <code>grid-area</code>
Alignment	<code>justify-items</code> , <code>align-items</code> , <code>place-items</code>
Individual alignment	<code>justify-self</code> , <code>align-self</code> , <code>place-self</code>

Whole grid alignment	<code>justify-content</code> , <code>align-content</code> , <code>place-content</code>
Animation	<code>animation</code>
Animation stages	<code>@keyframes</code>
Animation speed	<code>animation-timing-function</code>
Animation repetition	<code>animation-iteration-count</code>
Animation direction	<code>animation-direction</code>
Animation delay	<code>animation-delay</code>
Animation pause	<code>animation-play-state</code>
Animation final state	<code>animation-fill-mode</code>
Responsive condition	<code>@media</code>
Width condition	<code>min-width</code> , <code>max-width</code>

Responsive condition	@media
Width condition	min-width , max-width
Orientation	orientation
User motion preference	prefers-reduced-motion
User theme	prefers-color-scheme
Pointer	pointer , any-pointer
Hover capability	hover , any-hover

42. इस पूरे CSS topic का Master List

Category	सभी Important Properties / Methods	
Grid Basic	<code>display:grid</code>	
Grid Tracks	<code>grid-template-rows</code> , <code>grid-template-columns</code> , <code>grid-template</code> , <code>grid-auto-rows</code> , <code>grid-auto-columns</code> , <code>grid-auto-flow</code>	
Grid Functions	<code>repeat()</code> , <code>minmax()</code> , <code>fit-content()</code> , <code>subgrid</code>	
Grid Units	<code>fr</code> , <code>%</code> , <code>px</code> , <code>auto</code> , <code>min-content</code> , <code>max-content</code>	
Grid Placement	<code>grid-row</code> , <code>grid-row-start</code> , <code>grid-row-end</code> , <code>grid-column</code> , <code>grid-column-start</code> , <code>grid-column-end</code> , <code>grid-area</code> , <code>span</code>	
Grid Areas	<code>grid-template-areas</code> , <code>grid-area</code>	
Grid Gap	<code>gap</code> , <code>row-gap</code> , <code>column-gap</code>	
Grid Item Alignment	<code>justify-self</code>  <code>align-self</code> , <code>place-self</code>	

Grid Item Alignment	<code>justify-self</code> , <code>align-self</code> , <code>place-self</code>
Grid Cell Alignment	<code>justify-items</code> , <code>align-items</code> , <code>place-items</code>
Grid Container Alignment	<code>justify-content</code> , <code>align-content</code> , <code>place-content</code>
Animation Basic	<code>animation</code> , <code>animation-name</code> , <code>animation-duration</code> , <code>animation-delay</code>
Animation Speed	<code>animation-timing-function</code> , <code>linear</code> , <code>ease</code> , <code>ease-in</code> , <code>ease-out</code> , <code>ease-in-out</code> , <code>steps()</code> , <code>cubic-bezier()</code>
Animation Repeat	<code>animation-iteration-count</code> , <code>infinite</code>
Animation Direction	<code>animation-direction</code> , <code>normal</code> , <code>reverse</code> , <code>alternate</code> , <code>alternate-reverse</code>
Animation State	<code>animation-play-state</code> , <code>running</code> , <code>paused</code>
Animation End State	<code>animation-fill-mode</code> , <code>none</code> , <code>forwards</code> , <code>backwards</code> , <code>both</code>
Animation Keyframes	<code>@keyframes</code> , <code>from</code> , <code>to</code> , <code>0%</code> , <code>25%</code> , <code>50%</code> , <code>75%</code> , <code>100%</code>

Animation Keyframes	@keyframes , from , to , 0% , 25% , 50% , 75% , 100%
Advanced Animation	animation-timeline , animation-range , animation-composition
Media Query	@media
Width Queries	width , min-width , max-width
Height Queries	height , min-height , max-height
Screen	orientation , aspect-ratio , resolution
Color/Display	color , monochrome , color-index , display-mode
Input	hover , any-hover , pointer , any-pointer
User Preference	prefers-color-scheme , prefers-reduced-motion , prefers-contrast , forced-colors , inverted-colors
Media Operators	and , not , comma/OR conditions
Responsive Units	% , vw , vh , vmin , vmax , rem , em , ch

